

Module 7: Scenario-Based Fault Exercises

- PLC Tabletop Training Program · Lab / Assessment · 4 h
- v1.0 (April 2026)

Learning objectives

- Independently diagnose and resolve a tripped area scanner fault
- Identify and correct an IP connectivity failure
- Trace a missing sensor signal from PLC to field device
- Identify a logic fault (incorrect timer preset) by tracing the ladder rung
- Replace a faulty push button and verify proper operation

Overview

- Five realistic fault scenarios pre-staged on the bench. Diagnose and resolve each independently using the skills from Modules 1–6.

Lesson: How Scenarios Work

- Structure, expectations, and grading criteria for fault exercises

How Scenarios Work

- Each scenario is pre-staged on the bench by the instructor. You receive only a symptom description — no hints about the cause. Your job is to diagnose and resolve the fault independently using the tools and methods from Modules 1–6.

Exercise Structure

- Table: Phase | Duration | Activity
- Briefing | 5 min | Trainer describes the symptom. You discuss your troubleshooting plan before touching anything.
- Execution | 25–30 min | Work independently (or in pairs) on the bench. Trainer observes but doesn't assist unless safety is at risk.
- Debrief | 10 min | Group discussion: What did you find? What steps did you take? Trainer reveals the planted fault and the ideal path.
- [INFO] Success Criteria: You must resolve each scenario within the time limit. Assessment looks at: correct fault identification, logical troubleshooting sequence (not random guessing), proper tool usage, and safe work practices throughout.